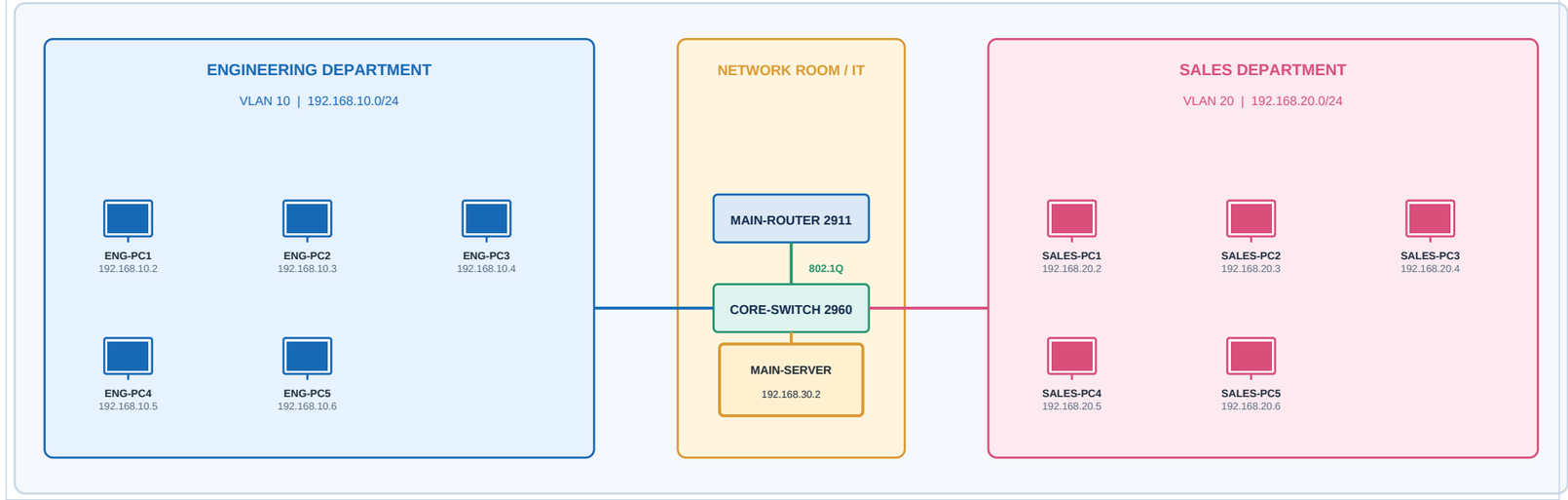
SMALL OFFICE NETWORK DESIGN USING CISCO PACKET TRACER

Engineering and Sales Departments with Shared Local Server (5 PCs Each)

OFFICE LAYOUT (INFRASTRUCTURE VIEW)



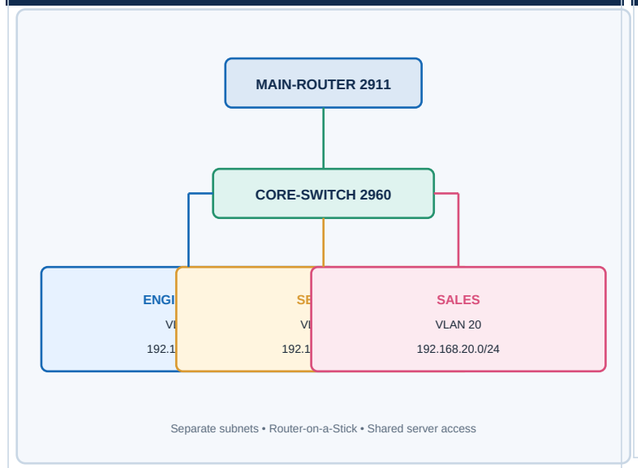
IP ADDRESSING PLAN

DEPARTMENT / DEVICE	VLAN	NETWORK ADDRESS	DEFAULT GATEWAY	HOST IP RANGE
Engineering (5 PCs)	10	192.168.10.0/24	192.168.10.1	192.168.10.2–.6
Sales (5 PCs)	20	192.168.20.0/24	192.168.20.1	192.168.20.2–.6
Server (1)	30	192.168.30.0/24	192.168.30.1	192.168.30.2

HARDWARE AND SOFTWARE REQUIREMENTS

HARDWARE & SOFTWARE
Cisco 2911 Router × 1
Cisco 2960 Switch × 1
Engineering PCs × 5
Sales PCs × 5
Main Server × 1
Copper straight-through cables
802.1Q trunk link
Cisco Packet Tracer

LOGICAL NETWORK TOPOLOGY (CISCO PACKET TRACER)



OBJECTIVES AND EXPECTED OUTCOME

OBJECTIVES	EXPECTED OUTCOMES
1. Design a small office LAN in Packet Tracer. 2. Separate Engineering and Sales using VLANs. 3. Create a dedicated server VLAN. 4. Configure inter-VLAN routing on Cisco 2911. 5. Verify PC and server connectivity.	✓ Separate departmental subnets. ✓ Engineering and Sales can communicate. ✓ Both departments can access the shared server. ✓ Network is organized and manageable. ✓ Connectivity verified using ping.

VLAN CONFIGURATION (CISCO 2960 SWITCH)

```
Switch> enable
Switch# configure terminal
hostname Core-Switch
vlan 10
 name ENGINEERING
vlan 20
 name SALES
vlan 30
 name SERVER
!
int range fa0/1-5
 switchport mode access
 switchport access vlan 10
int range fa0/6-10
 switchport mode access
 switchport access vlan 20
int fa0/11
 switchport mode access
 switchport access vlan 30
int g0/1
 switchport mode trunk
 switchport trunk allowed vlan 10,20,30
end
write memory
```

ROUTER CONFIGURATION (CISCO 2911)

```
Router> enable
Router# configure terminal
hostname Main-Router
int g0/0
 no shutdown
int g0/0.10
 encapsulation dot1Q 10
 ip address 192.168.10.1 255.255.255.0
int g0/0.20
 encapsulation dot1Q 20
 ip address 192.168.20.1 255.255.255.0
int g0/0.30
 encapsulation dot1Q 30
 ip address 192.168.30.1 255.255.255.0
end
write memory
```

VERIFICATION (PING TEST RESULTS)

```
ENG-PC1> ping 192.168.20.2
Reply from 192.168.20.2
Reply from 192.168.20.2
Reply from 192.168.20.2
Reply from 192.168.20.2
Packets: Sent = 4, Received = 4, Lost = 0

ENG-PC1> ping 192.168.30.2
Packets: Sent = 4, Received = 4, Lost = 0

SALES-PC1> ping 192.168.10.2
Packets: Sent = 4, Received = 4, Lost = 0
```

DEVICE IP DETAILS: Engineering PCs = 192.168.10.2–192.168.10.6 | Sales PCs = 192.168.20.2–192.168.20.6 | Main-Server = 192.168.30.2 | Subnet mask = 255.255.255.0 | Router gateways = .1 in each subnet. **PORT PLAN:** Engineering Fa0/1–5, Sales Fa0/6–10, Server Fa0/11, trunk G0/1; router subinterfaces on G0/0.